ComfortRoute – Iteration 3 Deliverable

Course: CSE 3311 – Software Engineering

Project: ComfortRoute – Transit Planning System

Iteration: iteration 3

Team: Team 3

Version: v0.3-iteration

Repository: https://github.com/Kass-ii/CSE-3311-

Iteration 3 Overview (out and back trips)

Iteration 3 focuses on supporting **origin-destination**, where a user can travel from an origin station to another station while minimizing time spent waiting outdoors

This iteration expands the system from simple routing to a more flexible structure while improving usability through map interaction, customization settings, and visualization features.

The system now allows users to:

- Select the origin and destination stations.
- Views route legs and timing.
- Use map interaction to plan a trip.
- Consider comfort factors such as indoor lobbies and station hours.

Design Rational

Earlier iteration intentionally restricted the routing problem:

- **Iteration 1**: same-line-backtracking.
- **Iteration 2**: Out-and-back of the same origin.
- **Iteration 3**: Origin-destination trips with improved UI and interaction.

These constraints were intentional design decisions to reduce technical risks and validate routing behavior step by step.

The final iteration will remove these constraints and support:

- Multi-line routing.
- Transfers.
- Extended detours.

Project Plan

Iteration Plan

- **Iteration 1:** Backtracking on same rail line(completed).
- **Iteration 2:** Out-and-back rail trips of the same origin (completed).
- **Iteration 3:** Out-and-Back trips (current iteration).
- **Iteration 4**: Multi-line trips, deployment, UI refinement.

Iteration 4 Plan

The final iteration will extend routing capabilities to:

Must have:

- Configure and deploy app in production environment
- Visual representation of suggested trip
- Multi line transfers

Should have:

- Bookmark frequently used stations
- Configurable transfer buffer

Could have:

- Scrape DART service alerts
- Multiple trip options to compare

Input and Outputs

Planner Input:

- Origin station.
- Origin destination.
- Departure time.
- User settings.

Planner Outputs:

- Route summary
- Route legs.
- Travel time and wait time.
- Indoor lobby availability.
- Station hours.

Map Inputs:

- User location.
- Station data.
- Settings.

Map Outputs:

- Nearby stations.
- Station markers and popups.
- Selected origin/destination.
- Colored route lines.

Use Case: Plan an Out-and-Back Trip

- User opens the map page.
- User selects a station as an origin.
- User selects another station as destination.
- User opens the planner.
- User inputs departure time.

- System computes route legs.
- User views trips including timing and comfort details.

User Feedback

We reached out to friends who frequently use public transit and gave a demo of the app to get feedback. From these demos determined next steps to improve user satisfaction:

- Graphical representation of user trip
- bookmark favorite stations and save to user account
- show multiple trip options and compare comfortable trip with the fastest
- configurable buffer time between transfers
- A now button next to the departure time, replace -- with 00 if user leaves blank
- Display DART service alerts to user if they affect the route
- Dropdown menu to autocomplete station names in the planner

Risk

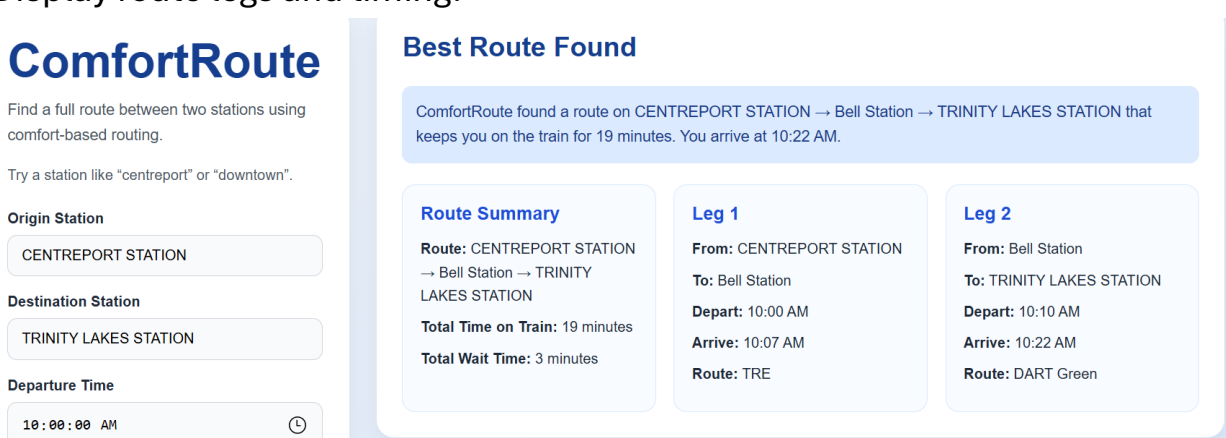
Risk	Time Impact	Probability	Risk Exposure
(1) Train platforms are far apart and transfers take longer than expected.	4 hours	10%	24 min
(2) Unable to deploy app to host	6 hours	20%	2 hours
(3) GTFS schedules do not reflect reality on transit	4 hours	60%	2.5 hours

1. Make transfer time user configurable
2. Use old raspberry pi as stopgap server
3. Display service alerts to users, filter alerts for those relevant to the user's trip and warn if route is affected.

Implemented Features

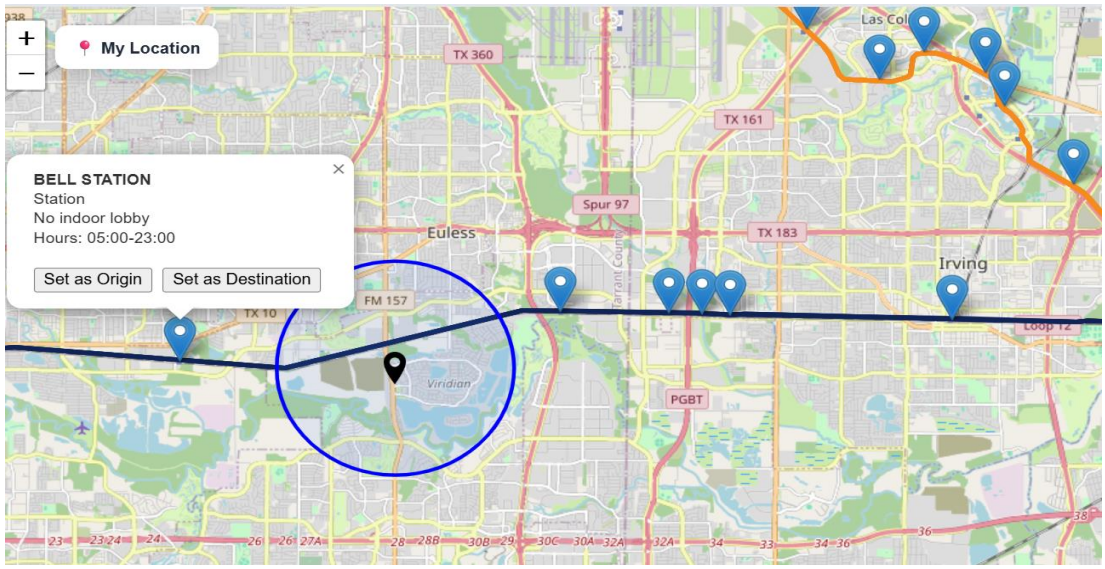
Routing:

- Planner supports origin — destination trips.
- Display route legs and timing.



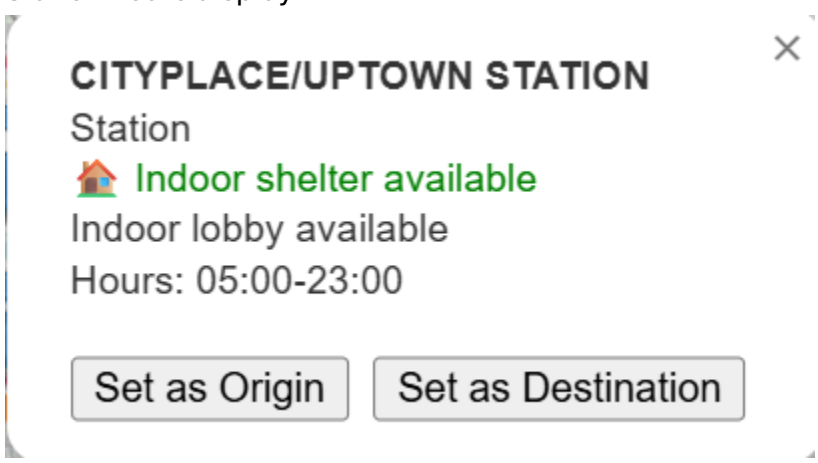
Map interaction:

- Select origin and destination from map.
- Show nearby stations.
- Display user location.



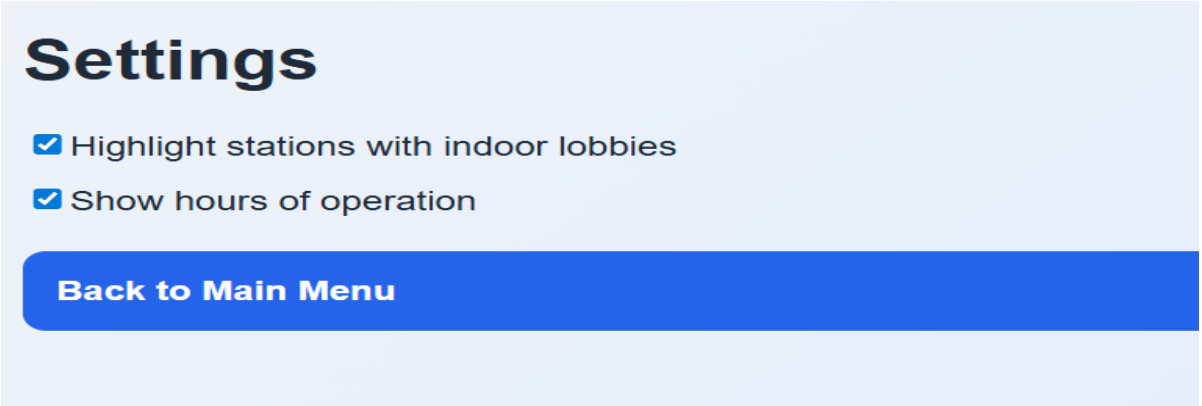
Comfort Features:

- Indoor lobby highlighting.
- Station hours display.



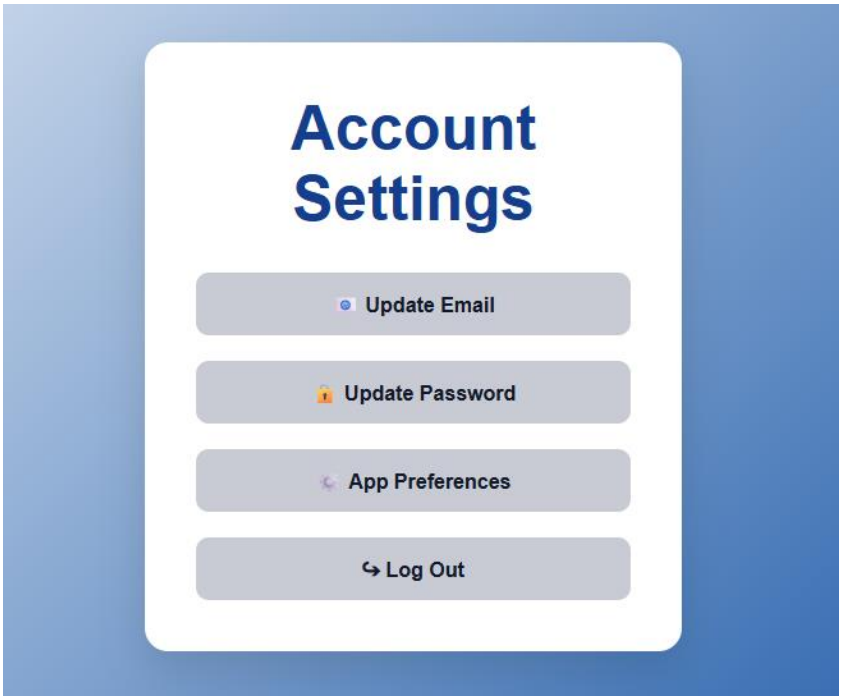
App Settings:

- Toggle indoor lobby visibility.
- Toggle station hours.



Account Settings:

- Update email address
- Update password
- Log-out system



Visualization:

- Colored route lines.
- Improved layout and user interface.

Work Completed (iteration 1-3)

- **Iteration 1**
    - Implemented out & back routing algorithm
    - Initialized GTFS data into a data structure
- **Iteration 2**
    - Created functional user interface
    - Created an interactive map of stations
    - Developed a planner for users to input their desired times and started stations
    - Upgraded initial algorithm
- **Iteration 3**
    - Implementation of out-and-back trips
    - Planner supports route paths from origin station to destination
    - Improved map interactivity
    - Added customizable UI settings
    - Added account settings and logout system

Conclusion

Iteration 3 offers more flexibility by allowing users to select their origin and desired destinations within inputted time constraints. Improved map interactivity allows for more sophisticated trip planning, while the addition of settings gives users the option to make personalized changes to the UI and their account.